

CSE 565 Performance Testing Project

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CSE 565: Software Verification and Validation

Performance Testing Project

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1 Introduction

This draft documents the performance-testing framework, the three baseline test cases, and the raw execution results. The analysis, conclusions, and improved stress-test strategy will be added after review of the Locust reports.

2 Task 1: Identification of the Performance Testing Framework

Locust 2.46.6 was selected as the performance-testing framework. Locust is a Python framework in which a user class defines the work performed by a simulated user. It records request count, failures, response times, response-time percentiles, and requests per second (RPS). The framework can be run from the command line in headless mode and can create HTML and CSV reports.

The official Locust training module used for the framework setup is the “Your first test” quick-start guide: <https://docs.locust.io/en/stable/quickstart.html>. The guide demonstrates defining a user class in `locustfile.py`, starting Locust, and reviewing response-time and RPS information. The official headless-mode guide used for the command-line test execution is <https://docs.locust.io/en/stable/running-without-web-ui.html>.

The instructor-provided `load_test_tasks.py` file was preserved without modification. The separate `locustfile.py` defines a Locust user workflow with the same six workload categories: data processing, simulated database query, API request, file I/O, computation, and logging. A controlled local profile was used so that the 10,000-user test could be executed without directing high-volume traffic to the public API in the starter script.

3 Task 2: Test Case Design

Each Locust user repeatedly executes the six workload categories in one workflow. Locust records a separate **TASK** result for each category. The controlled API task returns a simulated response; the remaining operations perform local processing, waiting, file I/O, computation, and logging. The three baseline cases use the same standard profile and a 30-second observation window.

Table 1: Baseline Locust test cases.

Test	Load level	Users	Spawn rate	Duration	Profile
Test 1	Baseline load	100	20 users/s	30 s	Standard
Test 2	Medium load	1,000	200 users/s	30 s	Standard
Test 3	High load	10,000	1,000 users/s	30 s	Standard

The following command is implemented by `run_locust_scenarios.sh`. It invokes Locust in headless mode and saves an HTML report, statistics CSV, statistics-history CSV, failures CSV, exceptions CSV, and console log for each load level.

```
CSE565_PROFILE=standard CSE565_API_MODE=simulated
CSE565_DURATION_SECONDS=30 .venv/bin/locust -f locustfile.py --headless
--users 100 --spawn-rate 20 --stop-timeout 15
--csv locust_results/users-100/locust
--html locust_results/users-100/locust_report.html
```

4 Task 3: Test Execution Results

The three baseline tests completed through Locust on October 4, 2026. Table 2 reproduces the aggregate row from each Locust `locust_stats.csv` report. Response-time values are in milliseconds. The result files are retained in the project under `locust_results/20261004-123006/`.

Table 2: Aggregate Locust results for the three baseline tests.

Users	Requests	Failures	Average (ms)	Median (ms)	Maximum (ms)	RPS
100	21,512	0	2.917	0.009	22.343	740.632
1,000	221,958	0	2.705	0.009	36.228	7,390.853
10,000	742,403	0	141.188	0.009	734.794	22,505.491

Table 3: Per-task Locust response-time results.

Load and task	Average (ms)	Median (ms)	RPS
100: Data Processing	0.193	0.058	123.462
100: Database Query	5.641	5.010	123.462
100: API Request	10.916	11.000	123.427
100: File I/O	0.554	0.082	123.427
100: Computation	0.029	0.009	123.427
100: Logging	0.169	0.022	123.427
1,000: Data Processing	0.082	0.057	1,232.608
1,000: Database Query	5.365	5.002	1,231.809
1,000: API Request	10.583	10.002	1,231.609
1,000: File I/O	0.147	0.074	1,231.609
1,000: Computation	0.012	0.009	1,231.609
1,000: Logging	0.041	0.021	1,231.609
10,000: Data Processing	0.063	0.057	3,812.246
10,000: Database Query	418.040	470.000	3,788.025
10,000: API Request	427.518	480.000	3,726.305
10,000: File I/O	0.127	0.072	3,726.305
10,000: Computation	0.010	0.009	3,726.305
10,000: Logging	0.037	0.021	3,726.305

References

- Locust. (n.d.). *Your first test*. Locust documentation. Retrieved October 4, 2026, from <https://docs.locust.io/en/stable/quickstart.html>
- Locust. (n.d.). *Running without the web UI*. Locust documentation. Retrieved October 4, 2026, from <https://docs.locust.io/en/stable/running-without-web-ui.html>